Enter Your Shopper ID?
SNAP, Social Stigma and Loyalty Programs

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Abstract: This paper considers one dimension along which social stigma may affect a SNAP recipient's shopping behavior: how the use of SNAP benefits impacts a shopper's decision to reveal their loyalty card number. To do this, we analyze over thirty months of transaction-level data from a large grocery store chain, covering over a hundred million transactions with over six million involving the shopper claiming SNAP benefits. Using within-household, across-transaction variation for identification, this paper finds an economically and statistically significant decrease in shoppers revealing their loyalty card number when using SNAP benefits, and that the decrease is significantly greater when the shopper goes through a cashier-operated checkout lane rather than a self-checkout lane. This decrease demonstrates another manner by which the social stigma associated with SNAP redemption imposes costs since not revealing causes the SNAP recipient to forego earning loyalty card credits which decreases the amount of discounts received in future transactions.

Keywords: SNAP; Social Stigma; Loyalty Program

Acknowledgments: Special thanks to Michigan State University and specifically to the Institute for Cyber-Enabled Research for providing the necessary computational resources. I would also like to thank the grocery store chain for the data and providing details not only on the data structure but also the SNAP benefit and loyalty program environments.

1. Introduction

The Supplemental Nutrition Assistance Program, or SNAP, is one of the largest social support programs in the US, distributing about $7.6 billion to 41.7 million people each month, as of FY2024.[1] While this means that SNAP participants make up about one percent of the US population, the take up rate of these benefits is not universal; as of 2022, the most recent year for which the USDA published this statistic, approximately 4 million people qualify but do not enroll for SNAP benefits.[2] As discussed in Ko and Moffitt (2022) this shortfall in uptake can occur for a variety of reasons, including transaction costs, difficulty in applying for the program, and social stigma. This stigma can manifest in other ways, from using SNAP at a different store than usual or going to the store at a different time than usual, as discussed in Rogers-Dillon (1995). However, Brumbaugh and Rosa (2009) describe how a great deal of shopping habit-derived stigma comes from face-to-face interactions with the cashier, so shoppers might choose to use self-checkout lanes instead of cashier-operated lanes when using SNAP[3]. Pukelis et al. (2026) details how that stigma interacts with online shopping, another avenue of purchasing without a face-to-face interaction with a cashier. The point of many of these manifestations is to allow the person to avoid associating the use of SNAP with themselves, as described in Ko and Moffitt (2022).[4]

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[1] https://www.ers.usda.gov/topics/food-nutrition-assistance/supplemental-nutrition-assistance-program-snap/key-statistics-and-research

[2] https://fna-bwbufwdzbabpezgc.z01.azurefd.us/sites/default/files/resource-files/ops-snap-trendsfy20-fy22-report.pdf

[3] https://www.feedingmifamilies.org/

[4] Stigma associated with government programs in the United States is outlined in Stuber and Schlesinger (2006) and Stuber and Kronebusch (2004), while stigma food stamps is documented in Eurofound (2015). Hatton, et al. (2024) and Seccombe, et al. (1998) provide much more detail about the social stigma associated with SNAP in particular. Daponte, Sanders and Taylor (1999) provide experimental evidence that lack of program knowledge contributes to food stamp non-participation while Gundersen and Oliveira (2001) provide evidence of selection issues associated with Food Stamp Program participation.

This paper contributes to this research by considering another dimension along which social stigma may affect a SNAP recipient's shopping behavior: how the use of SNAP benefits impacts a loyalty program member's decision to reveal their loyalty card number when shopping at a particular large grocery store chain. If social stigma is an issue, one would expect a loyalty program member to be less likely to provide her loyalty number on transactions involving SNAP benefits. The reason is that in order to sign up for the loyalty program, the shopper must submit their name, email address, phone number, and zip code directly to the retailer, either through the store's website or at an in-store customer service desk. For both cashier-operated and self-checkout lanes, the shopper reveals her loyalty program membership by entering her phone number on a touchscreen or scanning a bar code, and then entering a four-digit PIN. Shoppers may perceive that providing their loyalty card number allows the store or even the cashier to identify them more easily, and, if using SNAP benefits, map those benefits to the shopper. This could increase the stigma cost when SNAP benefits are claimed for that transaction. If they don't provide the loyalty number, most shoppers are likely unaware that the chain may still be able to identify them primarily through their credit card number. Each purchase where the shopper discloses her loyalty card number earns "credits", which can be redeemed for discounts on transactions.[5] This disclosure is entirely voluntary; the shopper can always just not input her loyalty program account number, and thereby not gain the benefits, but also avoid any perceived association between her identity and the transaction. In addition, the stigma cost a shopper incurs from this perceived association may depend on whether there is a cashier in the checkout lane or whether the shopper uses a self-checkout lane.

[5] It is important to note that these discounts are generally applied to future transactions, and are usually not discounts associated with specific products.

We analyze thirty months of transaction-level data from a large grocery store chain, covering over a 100 million transactions, to examine two primary research questions. First, how does the redemption of SNAP benefits affect whether shoppers provide their loyalty card number? And, second, does the decision to provide this number vary based on whether the shopper uses a cashier-operated checkout and self-checkout lanes? If stigma is an important factor, we would expect that the use of SNAP benefits would reduce the rate of loyalty program disclosure, and that the reduction would be more at cashier-operated points of sale than at self-checkout lanes. Using within-household, across-transaction variation for identification, the estimates from our two-way fixed effects model indicate that the probability a shopper provides their loyalty card number decreases by 0.015 when the transaction involves SNAP benefits. Given that the probability loyalty program members, who have ever used SNAP benefits in a transaction, do not reveal their loyalty card number is approximately 0.28, this represents over a five percent increase of not revealing. Estimating a similar specification which allows this increase to vary based on type of checkout lane, we find the increase in the probability of not revealing, when SNAP benefits are claimed in the transaction, is 0.035 greater when the shopper goes through a cashier-operated lane rather than a self-checkout lane. These two main empirical results provide further evidence of the social stigma associated with SNAP benefits and illustrates another manner by which this stigma is manifested.

2. Data and Descriptive Statistics

We use a data set from a large grocery store chain that contains information on all transactions between December 2022 and May 2025. The transaction-level variables include the date and time, store identification number, type of checkout lane, the value of the transaction,

whether SNAP benefits were claimed, and the value of the SNAP benefits. They also include household identifiers for almost a million different households and the loyalty card number if provided by the shopper.[6,7]

Table 1 contains summary statistics of our data at both the transaction and household level. Our data include slightly over a hundred million transactions from almost a million households. In terms of transactions, the shopper revealed her loyalty card number in almost three quarters of the 105 million transactions and less than six percent of these transactions involved SNAP benefits. The average expenditure per transaction is $53.40 with the average number of transactions for a household being slightly over four per month and households averaging a monthly expenditure of $218. We classify three types of checkout lanes: cashier-operated lanes, self-checkout lanes and departmental lanes (e.g. pharmacy, garden center, bakery, etc.).[8] The fraction of transactions that took place at self-checkout lanes was 0.636, compared to 0.157 using cashier-operated lanes.[9]

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[6] We drop observations if there was no household identifier, as well as household observations who had never claimed SNAP benefits, who had never provided a loyalty card number (i.e., unlikely participate in the loyalty club program), and who had claimed WIC benefits during the 30 months. Observations are also dropped if the transactions were on-line (delivery and pickup orders) and if all items in a transaction are covered by SNAP benefits. If a shopper purchases only items covered by SNAP on a shopping trip, the loyalty card number is automatically identified in the data.

[7] The reason we drop all transactions for households that have ever claimed WIC benefits, and do not test whether social stigma affects a shopper's decision to provide her loyalty number when claiming WIC benefits, is because of our concern that a WIC shopper's household identifier may not be the same across transactions. Qualifying for and claiming WIC benefits is often correlated with life changes (such as how a shopper pays for groceries, a shopper's address and perhaps even a shopper's name) that would result in the grocery store chain altering her household identifier. This would likely bias the estimates from our primary empirical specifications which use within households, across transaction variation for identification.

[8] Both the cashier-operated and self-checkout lanes are at the front of the grocery store. The departmental lane designation also include other lane types such as alcohol only and service desk checkouts.

[9] The average household used self-checkout lanes for 60.9 percent of their transactions, and cashier-operated checkouts for 22.7 percent. The fact 0.227 is more than 0.157 means that households with fewer transactions were more likely to go to the cashier-operated checkout.

Table 1: General Descriptive Statistics

Transaction-Level Data		Household-Level Data	
Fraction where loyalty card number provided	0.720		
Fraction of Transactions with SNAP benefits claimed	0.058	Average Transactions per Household per Month	4.07 (6.27)
Average Expenditure	$53.40 ($64.59)	Average Monthly Expenditure	$218 ($304)
Fraction used Cashier-operated Checkout	0.157	Fraction used Cashier-operated Checkout	0.227
Fraction used Self-Checkout	0.636	Fraction used Self-Checkout	0.609
Fraction used Departmental Checkout	0.207	Fraction used Departmental Checkout	0.164
Total Number of Transactions	104,916,703	Total Number of Households	857,566

While the descriptives in Table 1 present several variables that, along with SNAP benefit claims, may influence a shopper's decision to provide a loyalty card number, it does not provide insight on how this decision varies across time and season. Figure 1 depicts the proportion of monthly transactions where the shopper revealed her loyalty number and the proportion where SNAP benefits were not claimed. The graph indicates an increase across time in the probability a shopper reveals her loyalty card and, besides at the start of 2023, a relatively constant proportion of transaction involving SNAP benefits.

Figure 1
Proportion of Transactions with Loyalty Card Number Provided and SNAP Benefits Not Claimed by Month

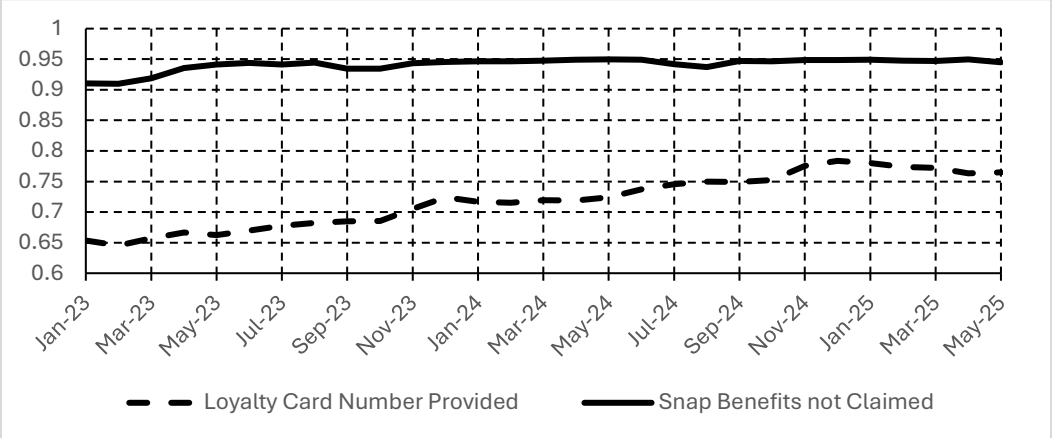


Table 2 provides a description of the transaction level data, categorized by whether or not the household ever used SNAP benefits, and for those who have (as in Table 1), whether or not they used SNAP benefits on the transaction.[10] Of note are three statistics. First, for households who have ever used SNAP benefits, the likelihood of a shopper providing a loyalty card number for the transaction is higher when using SNAP benefits (0.766) than when not using SNAP benefits (0.717). Households which have never used SNAP are less likely (0.676). Secondly, households who have ever used SNAP benefits made much larger transactions when using SNAP benefits, with an average transaction value of $103.59, than when not using SNAP benefits, with an average transaction value of $50.25. Third, we consider the checkout lane decision and note that for households who have ever used SNAP benefits, the probability of using a cashier-operated lane are significantly greater and the probability of using a self-checkout lane is only slightly greater for transactions involving SNAP benefits. While both are greater, due to the fact that almost all departmental lanes do not allow SNAP benefit claims, the ratio of self-checkout use to cashier-operated lane use is much lower when using SNAP benefits (with 0.680/0.316= 2.15 times more transactions at the self-checkout lane) than when not using SNAP benefits (with 0.633/0.147= 4.31 times more transactions at the self-checkout lane).

If social stigma reduced the probability a shopper provides her loyalty card number on a transaction using SNAP benefits, one would not expect the probability of providing to be greater when claiming SNAP benefits (0.766>0.717). However, it is important to note that average expenditure was over twice as much for SNAP benefit transactions ($103.59 compared to

[10] Table 2, along with Tables 1 and 3, only includes shoppers with household identifiers who are enrolled in the loyalty program. We infer their loyalty program membership based on at least one transaction where the shopper provided a loyalty card number. We expect that households who have never provided a loyalty card number over the thirty months are likely not to be program participants. The reason we restrict the analysis to these households is because we are interested in the shopper's decision to reveal their loyalty card number at checkout, and people who aren't enrolled in the loyalty program aren't making that choice.

$50.25) and the benefits associated with the loyalty program increase with expenditures. Interestingly, the proportion of shoppers using the cashier-operated lane, where stigma is perhaps a more important issue, is greater for transactions including SNAP benefits.

Table 2: Transaction Level

	Have Used SNAP		*Never Used SNAP*
	SNAP Benefits Used in Transaction	SNAP Benefits Not Used in Transaction	
Fraction loyalty number provided	0.766	0.717	0.676
Transaction Expenditure	$103.59	$50.25	$63.06
	($89.54)	($61.39)	($69.11)
Fraction Used Cashier-operated lane	0.316	0.147	0.200
Fraction Used Self Checkout lane	0.680	0.633	0.614
Fraction Used Departmental lane	0.004	0.220	0.186
Transactions	6,110,526	98,806,177	425,065,175

Table 3 focuses on checkout lanes by dividing transactions based on whether or not SNAP benefits were claimed and then breaking them further down by cashier-operated, self-checkout, and department lanes. The most interesting aspect of Table 3 is that the difference in the probability a shopper provides her loyalty number when not using SNAP benefits relative to the probability when using SNAP benefits in the transactions, is greater when the shopper selects a cashier-operated checkout lane. When the shopper selects a self-checkout lane, there is basically no difference in the probabilities. Specifically, this difference is 0.036 (0.801-0.765) for a cashier-operated checkout lane and -0.003 (0.764-0.767) for a self-checkout lane. If the increase in a shopper's stigma costs of providing her loyalty number when claiming SNAP benefits is greater with a cashier-operating checkout lane, relative to a self-checkout lane, we would expect the probability difference above to be greater when there is a cashier. Finally, Table 3 indicated that expenditures are greater for transactions using a cashier-operating checkout lane and transaction when SNAP benefits are claimed (irrespective of the type of checkout lane selected).

Table 3: Transactions by Checkout Lane

	SNAP Benefits Used			SNAP Benefits Not Used		
	Cashier-Operated	Self Checkout	Dept. Lane	Cashier-Operated	Self Checkout	Dept. Lane
Fraction Loyalty Number Provided	0.765	0.767	0.698	0.801	0.764	0.527
Transaction Expenditure	$153.52 ($110.18)	$80.38 ($66.22)	$98.12 ($100.72)	$109.74 ($96.15)	$44.43 ($49.56)	$27.36 ($30.00)
Transactions	1,933,510	4,153,134	23,882	14,528,657	62,576,124	21,701,396

3. Empirical Analysis

The descriptives in Table 2 indicate that there are significant differences across households depending on whether the household has ever used SNAP benefits. Not only does the number of transactions differ but also whether the shopper provides her loyalty card number, total expenditures, and whether cashier-operated, self or departmental checkout lane is used for the transaction. Both Tables 2 and 3 indicate that for households who have used SNAP benefits at the grocery chain in this 30-month period, there are significant differences across transactions in expenditures and type of checkout lane selected, depending on whether or not SNAP benefits are claimed in the transaction. The costs and benefits associated with a shopper providing her loyalty card number is likely to vary based on these transaction characteristics. In addition, we find that households that have used SNAP benefits are more likely to provide their loyalty card number for transactions where SNAP benefits are claimed relative to those transactions where SNAP benefits are not claimed. This could be attributable to minimal stigma associated with claiming SNAP benefits or SNAP benefit claims being correlated with other transaction characteristics that influence the decision to reveal their loyalty card number.

The descriptives in these tables and Figure 1, suggest that to credibly test if stigma influences whether a shopper reveals her loyalty number, we need to account for household

characteristics, time of transaction and other transaction characteristics (i.e., expenditures and type of checkout lane), along with SNAP benefit claims; all which influence the marginal benefit and/or marginal cost of this decision. We do this by estimating the following linear, two-way fixed effects specification with the dependent variable being an indicator for whether the loyalty card number is provided by household h for transaction tr ($LoyaltyUsed_{htr}$). [11]

$$LoyaltyUsed_{htr} = \alpha_h + \alpha_m + \boldsymbol{X_{tr}} \bullet \boldsymbol{\beta_{tr}} + \varepsilon_{htr} \qquad (1)$$

In terms of notation, α_h and α_m denote household and monthly fixed effects, respectively; the vector $\mathbf{X_{tr}}$ contains transaction characteristics which include a SNAP benefits indicator, transaction expenditure, and indicators for type of checkout lane; and ε_{htr} is an idiosyncratic error term that is allowed to be correlated across store observations. Household fixed effects are included because of the limited household information we observe and our expectation, in part based on the descriptive tables, that unobservable household characteristics are likely to significantly influence the decision to provide a loyalty card number. By including household fixed effects, the specification controls for unobservable household characteristics that do not vary with time, resulting in the coefficient estimates being identified by within household, across transaction variation in loyalty card number disclosure. Figure 1 depicts how this decision also varies over time, and this is why monthly fixed effects are also included as covariates. In addition to household and time, the shopper's decision to provide a loyalty card are likely influenced by the characteristics of the transaction which is why $\boldsymbol{X_{tr}}$ includes a number of these characteristics. The amount of transaction level discounts associated with loyalty card points

[11] We estimate a linear model instead of a non-linear model, such as a probit, because of the incidental parameters issue (resulting in inconsistent estimates) and the computational issues associated with having almost a million households.

increase with total expenditures so the benefit of providing loyalty card number increases with expenditures.[12] Along with the benefit, the cost of loyalty number disclosure is likely to vary based on lane type.

Table 4 contains the coefficient estimates from Equation (1). As noted in Table 1, the probability of a shopper not revealing her loyalty card number for a given transaction is 0.280. The estimates in the first column of Table 4, when the specification contains household fixed effects, indicate that claiming SNAP benefits increases the probability of not revealing by 0.015, which is a 5.4 percent increase (0.015/0.280). This effect is economically significant and statistically significant at the one percent level. In addition, the estimate of the transaction expenditure coefficient indicates that the probability of not revealing the loyalty card number decreases by 8.4 percent with every additional hundred dollars spent. This is expected; the more a shopper spends on a transaction, the more benefit she gains by revealing her loyalty number. Finally, when controlling for household fixed effects, the probability shoppers reveal their loyalty card numbers does not vary appreciably based on whether they use a cashier-operated lane compared to a self-checkout lane. The second column contains estimates when household fixed effects are not included in the Equation 1 specification. The main thing to note from this specification is the coefficient estimate associate with the SNAP benefit indicator suggests a significantly larger increase in the probability of a shopper not revealing their loyalty card number when the transaction involves SNAP benefits. The magnitude of this negative coefficient, when household fixed effects are not included, is likely partly attributable to

[12] Loyalty card points also vary based on the type of products purchased. While not presented in the tables, the types of products purchased differs appreciably for transaction where SNAP benefits are claimed compared to transaction where SNAP benefits are not used. It also varies across checkout lane type.

unobserved household characteristics that influence both the shopper's decision to reveal her loyalty card number and to use SNAP benefits.

Table 4:
Effect of Transaction Characteristic on Loyalty Number Disclosure

	Household fixed effects	No Household fixed effects
SNAP Benefits Indicator	-0.015**	-0.033**
	(0.001)	(0.002)
Transaction Expenditure (in $100s)	0.084**	0.078**
	(0.001)	(0.001)
Cashier-Operated Checkout Lane	0.225**	0.208**
	(0.003)	(0.004)
Self-Checkout Lane	0.228**	0.224**
	(0.003)	(0.004)
Monthly Fixed Effects	Yes	Yes
Household Fixed Effects	Yes	No
R-Squared	0.34	0.07
Observations	104,916,703	104,916,703

Robust Standard Errors in parentheses. Standard errors are clustered at the store level. **p<0.01

As conjectured in the introduction, the stigma cost associated with providing a loyalty number when using SNAP benefits may be greater when there is a grocery store employee at the checkout lane – relative to a self-checkout lane. The fact that, in the above specification, the coefficient estimates associated with cashier-operated and self-checkout lanes are almost identical does not suggest otherwise because it does not specifically consider those transactions with SNAP benefits claimed. To empirically test the conjecture, we interact cashier-operated checkout lane and self-checkout lane with the SNAP benefit indicator. We include these interaction terms as covariates in the prior specification and present the estimates in Table 5.

If the stigma associated with using SNAP benefits is greater for shoppers in a cashier-operated checkout lane and shoppers are less likely to enter their loyalty card number when the stigma is greater, we would expect disclosure to be less when using SNAP benefits at a cashier-

operated lane compared to a self-checkout lane.[13] This is exactly what we find with the negative coefficient estimate associated with the cashier-operated*SNAP interaction term having a larger magnitude than the coefficient estimate associated with the self-checkout*SNAP interaction term. These estimates suggest that when using SNAP benefits, the probability a shopper enters her loyalty card number is 0.035 (-0.145-(-0.180)) less when the shopper uses a cashier-operated lane compared to a self-checkout lane. This difference is statistically significant at the one percent level.

Table 5:
Heterogeneous Effects on Loyalty Number Disclosure
Based on Shopper Using Cashier- versus Self-Checkout Lane

	Household fixed effects	No Household fixed effects
SNAP Benefits Indicator	0.140**	0.125**
	(0.006)	(0.001)
Transaction Expenditure (in $100s)	0.084**	0.078**
	(0.001)	(0.001)
Front of Store Checkout Lane	0.225**	0.212**
	(0.003)	(0.004)
Self-Checkout Lane	0.228**	0.224**
	(0.003)	(0.004)
Cashier-operated Lane * SNAP Benefits	-0.180**	-0.190**
	(0.005)	(0.007)
Self- Checkout Lane * SNAP Benefits	-0.145**	-0.144**
	(0.006)	(0.007)
Monthly Fixed Effects	Yes	Yes
Household Fixed Effects	Yes	No
R-Squared	0.34	0.07
Observations	104,916,703	104,916,703

Robust Standard Errors in parentheses. Standard errors are clustered at the store level. **p<0.01

[13] We are uncertain what to expect when comparing departmental checkout lanes with cashier-operated or self-checkout lanes because departmental lanes are significantly different than the "front of the store" lanes: both cashier-operated and self-checkout.

4. Conclusion

It has been well-established that using the social safety net, including SNAP, carries a degree of stigma (Ko and Moffitt, 2022), and that this stigma reduces the take-up rate of SNAP benefits (Stuber and Schlesinger, 2006). There is also evidence that, along with participation decisions, this social stigma affects SNAP recipient behavior in other manners including whether to shop on-line or in-store, what store to shop at, and when to go shopping. This paper contributes to this research area by examining how a shopper's use of SNAP benefits impacts her decision to input her loyalty account number and whether this varies based on whether the shopper uses a cashier-operated or self-checkout lane. Our empirical analysis finds strong, credible evidence that the use of SNAP benefits increases the probability a shopper does not reveal her loyalty card by over five percent and that this increase in probability is greater when the shopper goes through a cashier-operated lane rather than a self-checkout lane. The reason we perceive these findings convincing is because the empirical specifications use within household, across transaction variation for identification.

While our ability to use within household variation for identification is the primary strength of our analysis, there are limitations. The primary limitations are the inability to address the selection issues associated with (i) a shopper decision to participate in the loyalty card program and (ii) the shopper deciding whether to use a cashier-operated or a self-checkout lane. The fact that we do not observe loyalty card disclosure for those shoppers who do not enroll in the program and are unable to credibly address the decision to enroll, makes us cautious when applying our empirical results to households who do not participate in the loyalty card program. It may be the case that the social stigma associated with using SNAP benefits differ between shoppers who participate in the loyalty card program and those who don't. Our data also does not

allow us to credibly address the selection issue associated with what type of checkout lane a shopper chooses. That said, we conjecture that the increase in stigma cost associated with using a cashier-operated checkout lane, rather than self-checkout lane, is greater for shoppers who experience a large stigma cost when claiming SNAP benefits on a transaction. This would cause shoppers with large stigma costs to more likely select a self-checkout lane and we would expect these shoppers to also be less likely to disclose their loyalty card number when claiming SNAP benefits. If this is indeed the case, our estimate of the decrease in probability of revealing the loyalty card number, when the shopper goes through a cashier-operated lane rather than a self-checkout lane, is likely biased toward zero causing our specification to underestimate the actual difference.

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